The feasible-normal condition is necessary within the monotone convex class

An exact-size convex coordinatewise-monotone test is admissible iff all its normals lie in $\mathcal{F}$.

Setup. $X \sim N_2(\mu, I_2)$, $\Lambda = \{\mu_2 \le \mu_1\}$, $\Lambda_0 = \{\mu \in \Lambda : \mu_1 \le 0\}$, $\Lambda_1 = \{\mu \in \Lambda : \mu_1 > 0\}$. A nonrandomized test is $\phi_A = \mathbf{1}\{x \notin A\}$ with acceptance region A; its power is $\beta_{\phi_A}(\mu) = P_\mu(X \notin A)$. Write

$$T_1(x) = x_1 + x_2, \qquad T_2(x) = -x_2, \qquad x \succeq_\Lambda y \iff T_1(x) \ge T_1(y) \text{ and } T_2(x) \ge T_2(y).$$

The positive cone of $\succeq_\Lambda$ is $P = \text{cone}\{(1,0),(1,-1)\} = \{d : d_2 \le 0,\ d_1 + d_2 \ge 0\}$ (so $x \succeq_\Lambda y \iff x - y \in P$). For $\mu \in \Lambda_1$ write $\mu = (u, u - v)$ with $u > 0, v \ge 0$; then, relative to P_0,

$$L_\mu(x) = \frac{dP_\mu}{dP_0}(x) = \exp\Big\{u\,T_1(x) + v\,T_2(x) - \tfrac{1}{2}(\mu_1^2 + \mu_2^2)\Big\}, \qquad (*)$$

which is strictly increasing in T_1 (since $u > 0$) and nondecreasing in T_2 (since $v \ge 0$). A test is *coordinatewise monotone* (M1) if A is a lower set; equivalently every supporting normal of a convex A is ≥ 0. Finally

$$\mathcal{F} = \{a : 0 \le a_2 \le a_1\} = \text{cone}\{(1,0),(1,1)\}.$$

We prove the converse of the feasible-normal sufficiency theorem, restricted to the M1 class.

Theorem 1. *Let $A \subseteq \mathbf{R}^2$ be closed, convex, and coordinatewise monotone, with $P_0(X \in A) = 1 - \alpha$. If A has a supporting normal a with $a_2 > a_1$, then ϕ_A is inadmissible: there is a closed convex coordinatewise-monotone test of exact size α whose power exceeds that of ϕ_A at* every *$\mu \in \Lambda_1$.*

Corollary 1. *Within closed convex coordinatewise-monotone exact-size tests, ϕ_A is admissible against all level-α tests if and only if every supporting normal of A lies in $\mathcal{F}$.*

The "if" direction is the feasible-normal sufficiency theorem. For "only if": within M1, monotonicity forces $a \ge 0$, so a normal outside $\mathcal{F}$ is a normal with $a_2 > a_1$, and Theorem 1 applies. □

The proof builds one explicit competitor. Let

$$\widehat{A} := \text{cl}\,\big(A - P\big), \qquad A' := \widehat{A} \cap \{x : T_1(x) \le b\},$$

with b calibrated below. We use three lemmas: $\widehat{A}$ is the smallest $\mathcal{F}$-normal set above A (Lemma 1); A' is a legitimate exact-size M1 competitor (Lemma 2); and the gain and loss regions of A' against A occupy disjoint, correctly ordered x_2-ranges (Lemma 3), which forces domination.

Lemma 1 (the $\mathcal{F}$-normal hull)**.** *$\widehat{A}$ is closed, convex, coordinatewise monotone, a $\succeq_\Lambda$-lower set, contains A, and every nonzero direction of finite support satisfies $a \in \mathcal{F}$ with $h_{\widehat{A}}(a) = h_A(a)$. If A has a supporting normal a with $a_2 > a_1$, then $A \subsetneq \widehat{A}$ and $P_0(\widehat{A}) > P_0(A)$.*

Proof. $A - P = A + (-P)$ with $-P = \text{cone}\{(-1,0),(-1,1)\}$, a sum of convex sets, hence convex; its closure $\widehat{A}$ is closed and convex. For the support function, $h_{\widehat{A}}(a) = h_A(a) + h_{-P}(a)$, where $h_{-P}(a) = 0$ if $a \in (-P)^*$ and $+\infty$ otherwise, and

$$(-P)^* = \{a : a \cdot (-1,0) \le 0,\ a \cdot (-1,1) \le 0\} = \{a : a_1 \ge 0,\ a_2 \le a_1\}.$$

Since A is M1, $h_A(a) < \infty$ implies $a \ge 0$. Hence $h_{\widehat{A}}(a) < \infty$ implies $a \in \{a \ge 0\} \cap \{a_1 \ge 0, a_2 \le a_1\} = \mathcal{F}$, and there $h_{\widehat{A}}(a) = h_A(a)$. All finite normals lying in $\mathcal{F} \subseteq \{a \ge 0\}$ makes $\widehat{A}$ a lower set (M1); lying in $\{a_1 \ge 0, a_2 \le a_1\}$ makes $\widehat{A}$ a $\succeq_\Lambda$-lower set. Clearly $A \subseteq \widehat{A}$ because $0 \in -P$.

If A has a supporting normal a with $a_2 > a_1$ then $h_A(a) < \infty$ but $a \notin \mathcal{F}$, so the half-space $\{x : a'x \le h_A(a)\}$ that supports A is *not* a supporting half-space of $\widehat{A}$; thus $\widehat{A}$ extends strictly beyond A across that face, a full-dimensional difference, so $A \subsetneq \widehat{A}$ and $P_0(\widehat{A}) > P_0(A)$. □

Lemma 2 (the competitor)**.** *Assume $P_0(A) = 1 - \alpha$ and A has a normal a with $a_2 > a_1$. There is a unique $b \in \mathbf{R}$ with $P_0\big(\widehat{A} \cap \{T_1 \le b\}\big) = 1 - \alpha$. The set $A' = \widehat{A} \cap \{T_1 \le b\}$ is closed, convex, coordinatewise monotone, has all supporting normals in $\mathcal{F}$, and $P_0(A') = 1 - \alpha$. Consequently $\phi_{A'}$ has level α, attained at $\mu = 0$.*

Proof. $t \mapsto P_0(\widehat{A} \cap \{T_1 \le t\})$ is continuous and strictly increasing from 0 to $P_0(\widehat{A})$, and by Lemma 1 $P_0(\widehat{A}) > 1 - \alpha$; hence a unique b gives value $1 - \alpha$. A' is an intersection of the convex M1 $\succeq_\Lambda$-lower set $\widehat{A}$ with the half-space $\{T_1 \le b\}$ (normal $(1,1) \in \mathcal{F}$), so it is closed, convex, M1, with normals in $\mathcal{F}$. Coordinatewise monotonicity makes $\mu = 0$ least favorable on Λ_0: for $\mu \le 0$, coupling $X = \mu + Z$ with $Z \sim N_2(0, I_2)$ gives $\mu + Z \le Z$, hence $P_\mu(X \notin A') \le P_0(Z \notin A') = \alpha$. So the size is α, attained at 0. □

Lemma 3 (slice separation)**.** *Let $G := A \cap \{T_1 > b\}$ (the region newly rejected by $\phi_{A'}$) and $\ell := A' \setminus A$ (newly accepted). Then $P_0(G) = P_0(\ell) > 0$, and there is a level s_+ such that*

$$G \subseteq \{x_2 < s_+\}, \qquad \ell \subseteq \{x_2 > s_+\}.$$

Consequently $g \succeq_\Lambda \ell'$, with $T_1(g) > T_1(\ell')$ strictly, for every $g \in G$ and $\ell' \in \ell$.

Proof. Because $P_0(A) = P_0(A') = 1 - \alpha$ and $A \subseteq \widehat{A}$ gives $A \setminus A' = A \cap \{T_1 > b\} = G$ and $A' \setminus A = \ell$, the two symmetric differences have equal null mass: $P_0(G) = P_0(\ell)$. Since A has a normal $a_2 > a_1$, $\widehat{A} \supsetneq A$ and (as shown below) $\sup_A T_1 > b$, so G is full-dimensional and $P_0(G) > 0$.

For $s \in \mathbf{R}$ let $\beta(s) = \sup\{x_1 : (x_1, s) \in A\}$ (with $\sup \varnothing = -\infty$). As the upper boundary of a convex set, β is concave; set $m(s) = \beta(s) + s$, the maximal value of T_1 on the horizontal slice $\{x_2 = s\} \cap A$. Then m is concave. A direct computation from $\widehat{A} = \text{cl}(A - P)$, $P = \text{cone}\{(1,0),(1,-1)\}$, gives the slice of $\widehat{A}$:

$$(x_1, s) \in \widehat{A} \iff x_1 \le M(s) - s, \qquad M(s) := \sup_{s' \le s} m(s')$$

(the running maximum of m); indeed $(x_1, s) = y - \alpha(1,0) - \tau(1,-1)$ with $\alpha, \tau \ge 0$ forces $y_2 = s - \tau \le s$ and $x_1 \le (y_1 + y_2) - s$, maximized at $M(s) - s$.

Now read off the two slices. On $\{x_2 = s\}$,

$$G_s = \{x_1 : (x_1, s) \in A,\ x_1 + s > b\} = (\, b - s,\ \beta(s)\,],$$

nonempty iff $\beta(s) > b - s$, i.e. $m(s) > b$. And, since $A' \subseteq \{T_1 \le b\}$,

$$\ell_s = \{x_1 : (x_1, s) \in \widehat{A} \setminus A,\ x_1 + s \le b\} = (\, \beta(s),\ \min(M(s) - s,\, b - s)\,],$$

nonempty iff $M(s) > \beta(s) + s = m(s)$ *and* $b - s > \beta(s)$, i.e. $M(s) > m(s)$ and $m(s) < b$.

So a slice carries gain iff $m(s) > b$, and carries loss iff $m(s) < b$ and $M(s) > m(s)$. Because m is concave with $\max m =: m^\star > b$ (else $\sup_A T_1 \le b$, forcing $\widehat{A} \setminus A = \varnothing$ by the slice formula, contradicting the bad normal), the set $\{s : m(s) > b\}$ is an interval (s_-, s_+) containing the peak. For $s \le s_{\text{peak}}$, m is nondecreasing so $M(s) = m(s)$ and there is no loss. For $s > s_+$, $m(s) < b < m^\star = M(s)$, so loss is possible. Hence

$$\{s :\ G_s \neq \varnothing\} \subseteq (s_-, s_+), \qquad \{s :\ \ell_s \neq \varnothing\} \subseteq (s_+, \infty).$$

Thus every $g \in G$ has $g_2 < s_+$ and every $\ell' \in \ell$ has $\ell'_2 > s_+$, giving $T_2(g) = -g_2 > -\ell'_2 = T_2(\ell')$. Moreover $T_1(g) > b \ge T_1(\ell')$. Therefore $g \succeq_\Lambda \ell'$ with strict T_1-inequality, for all $g \in G,\ \ell' \in \ell$. □

Proof of Theorem 1. By Lemma 2, $\phi_{A'}$ is a closed convex M1 test of exact size α, hence a legitimate level-α competitor. Its power minus that of ϕ_A at $\mu \in \Lambda_1$ is, since the tests differ only on $G \cup \ell$,

$$\beta_{\phi_{A'}}(\mu) - \beta_{\phi_A}(\mu) = P_\mu(G) - P_\mu(\ell) = \int_G L_\mu \, dP_0 - \int_\ell L_\mu \, dP_0.$$

Let $m = P_0(G) = P_0(\ell) > 0$. Multiplying by m and using the product measure $P_0|_G \otimes P_0|_\ell$,

$$m\big(P_\mu(G) - P_\mu(\ell)\big) = \int_G \int_\ell \big\{L_\mu(g) - L_\mu(\ell')\big\}\, dP_0(\ell')\, dP_0(g).$$

By Lemma 3, every pair has $g \succeq_\Lambda \ell'$ with $T_1(g) > T_1(\ell')$; by $(*)$ and $u > 0, v \ge 0$ this gives $L_\mu(g) > L_\mu(\ell')$ pointwise. The double integral is therefore strictly positive, so

$$\beta_{\phi_{A'}}(\mu) > \beta_{\phi_A}(\mu) \qquad \text{for every } \mu \in \Lambda_1.$$

Hence ϕ_A is inadmissible. □

Remark 1 (what s_+ is). s_+ is the x_2-level above which the slicewise maximum $m(s) = \max_{x_1} T_1$ on A has already crested below b. For a rectangle $A = \{x_1 \le c_1, x_2 \le c_2\}$ one has $\beta(s) = c_1$ for $s \le c_2$, so $m(s) = c_1 + s$ is increasing up to the corner and $s_+ = c_2$: the separation level is exactly the horizontal face, $G = \{x_2 \le c_2,\ T_1 > b\}$ and $\ell = \{x_2 > c_2,\ T_1 \le b\}$, and $A' = \{x_1 \le c_1,\ x_1 + x_2 \le b\}$ is the gated test. The theorem is that construction run on a general convex M1 body, with s_+ playing the role of the rectangle's second cutoff.

Remark 2 (scope). Only the maintained test is asked to be convex; coordinatewise monotonicity (M1) is what pins the size at $\mu = 0$, so that the rearrangement preserves level rather than merely preserving null mass at 0. Dropping M1 reopens the regime of normals with $a_2 < 0$ (tests decreasing in x_2), for which $\mu = 0$ need not be least favorable and the present argument does not apply.